

some hydrozoans, including the Portuguese man-of-war. This animal looks much like a jellyfish but is actually a colony of polyps, one of which forms a baglike float (see opposite). However, floating colonies are comparatively rare: many more invertebrates—including corals, bryozoans, and sea squirts—form fixed colonies, attached to all kinds of objects, from shells to the seabed. Individuals in such colonies are usually connected to one another through joined gut and nerve networks, which allow sensory signals to be passed from one individual to its neighbors.



CENTER OF ATTENTION

Surrounded by workers, a queen honey bee controls all the activity in her hive. She does this by giving off volatile hormones—called pheromones—that evaporate and spread throughout the nest.

Invertebrate colonies are less common on land, and the members of terrestrial colonies are separate, rather than joined. They reach their greatest development in social insects, which include termites, as well as ants, bees, and wasps. Despite their apparent independence, the members of these colonies rely on each other for survival, just as much as if they were fastened together. In each colony, one individual—the queen—normally lays all the eggs, while the role of the other members is to ensure that the colony is fed, housed, and defended from attack.

Feeding

Collectively, the world's invertebrates eat almost everything that is—or once was—alive. They include herbivores and carnivores, as well as a host of species that feed on dead remains. Their appetites embrace everything that we consume as food, as well as some highly specialized foodstuffs, such as rotting seaweed, feathers, fur, and even animal tears.

In water, a common feeding method among invertebrates involves filtering out edible matter, using body parts that act like a sieve. This way of life supports vast numbers of static invertebrates that live on the sea bed or near the shore. In these animals, evolution has crafted filtering apparatus out of a wide range of body parts: barnacles, for example, use their legs, while bivalves use their gills. Many of these filter feeders actively pump water through their filters, to increase the size of their catch. For these animals, pumping often has the added advantage that it supplies oxygen as well as food. Filter feeding is also used by many planktonic animals, including krill, which in turn provide food for baleen whales—the largest filter feeders of all. Underwater “grazing,” in which vegetation is scraped from a solid surface, is also common. It is a technique used by limpets, which have rows of microscopic teeth, and also by sea urchins, which have a set of 5 calcium carbonate jaws.

On land, there is no equivalent of filter feeding, although some web-spinning spiders come close to mirroring this way of life, using their webs to trap airborne animals. Instead, most terrestrial invertebrates have to actively forage for food. The smallest herbivores—including many wood-eating species—often burrow through their food, a lifestyle that helps protect them from attack, but most predatory invertebrates move around in the open. In

NATURE'S RECYCLERS

Dozens of dung beetles clamber over a pile of animal droppings. The beetles feed on the dung, and they also roll it away and bury it to provide food for their young.



herbivores and carnivores alike, mouthparts are often shaped to deal with particular sources of food. Caterpillars bite off and swallow their food in substantial chunks, but many other insects have a liquid diet. Moths and butterflies, for example, are equipped with a coiled proboscis or



GATHERING FOOD

Dwarfed by its load, a leafcutter ant carries food back to its nest. In some parts of the world, ants eat as much food as all the plant-eating vertebrates in their habitat.

“tongue” up to 12 in (30 cm) long, which can be unrolled to drink nectar from flowers or juices from fruit, while horse flies have a pair of blade-like jaws that can slice through skin, allowing them to lap up oozing blood. Despite being carnivorous, spiders are also liquid feeders. They have a tiny mouth, and feed by pouring digestive juices onto their prey or injecting it into its body, and then sucking up the slurry that results.

Given their immense numbers, many invertebrates have an important ecological impact as they feed.

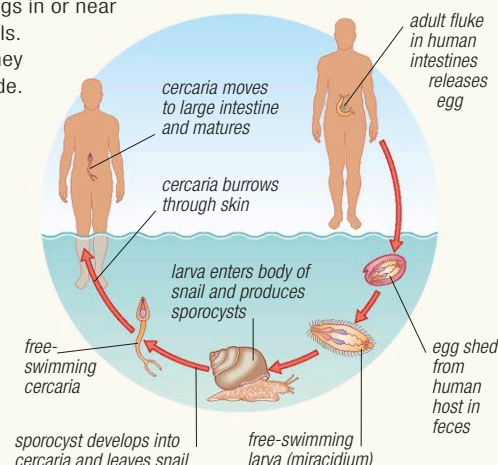
Earthworms play a key part in maintaining the fertility of soil, while insects have a major role in pollinating flowers and breaking down dead remains. In many habitats, invertebrates have a much greater effect than larger animals, but unless they are pests, their activity often goes unnoticed by humans.

PARASITES

Invertebrates make up the vast majority of the world's parasitic animals. Some—including ticks and leeches—live on the outside of their hosts. Many others, such as tapeworms and roundworms, live internally, feeding on their host's tissues or on the food that it consumes. External parasites can often survive with a single host, but many internal parasites have complex life cycles, involving several different hosts in sequence. These multiple hosts often include a vertebrate. Most parasites do not kill their hosts, but parasitoids are normally lethal. These animals, which include many insects, lay their eggs in or near the bodies of other animals. When the young hatch, they devour the host from inside.

LIFE CYCLE OF A BLOOD FLUKE

The human blood fluke is a parasitic flatworm that uses water snails as intermediate hosts. Its eggs reach water via untreated sewage, and produce free-swimming larvae. These multiply asexually inside snails, producing another swimming stage that attacks humans.



EATING AT LEISURE

Gripping the 2 halves of a mussel's shell, a starfish slowly pries them apart. It can evert its stomach through a paper-thin aperture between the mussel's shells to feed on it.

